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OS Assignment I

1)

→ An Operating System (OS) is system software that acts as an interface between the user and computer hardware. It manages resources such as the CPU, memory, files, input, output and applications. Different types of operating systems are designed according to the requirements of users and computing environments.

1) Batch Operating System

A Batch Operating System executes a collection of jobs in batches without requiring direct interaction from the user during execution. Similar jobs are grouped together and submitted to the system from processing.

Ex - An organization may process thousands of employee salary records together at the end of a month.

Advantages:

- Suitable for processing large volumes of similar tasks.
- Requires less user interaction.
- Efficient for repetitives jobs.

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Disadvantages
- Users cannot interact with a job while it is executing.
- A job may have to wait for previous jobs to complete.
- Debugging errors can be difficult.

2) Multiprogramming Operating System
A Multiprogramming OS allows several programs to be stored in the main memory at the same time. When one program is waiting for an input/output operation, the CPU can execute another program.
Ex- Program A is waiting for a file to be read from a disk, the CPU can execute Program B.

Advantages
- Improves CPU utilization.
- Reduces CPU idle time
- Allows multiple programs to remain in memory

Disadvantages
- Requires proper memory management.
- Scheduling becomes more complex.
- Memory and CPU resources must be carefully managed

3) Multitasking OS

A Multitasking OS allows multiple tasks or applications to run apparently at the same time. The operating system rapidly switches the CPU between different tasks.

Ex- A user can listen to music, browse the internet, and work on a document at the same time.

4) Time-Sharing OS

A Time-Sharing OS allows multiple users or processes to share the CPU. The CPU assigns a small amount of time, called a time slice, to each process.

After the time slice expires, the CPU can switch to another process.

5) Distributed OS

A Distributed OS manages a group of interconnected computers and coordinates their resources so that they can work together as a system.

The computers may be located at different physical location but communicate through a network.

6) Real-Time OS

A Real-Time OS (RTOS) is designed to respond to events and complete tasks within a specified time limit. It is used where timing is critical.

Q.2)

→ A process is a program that is currently being executed. A program stored on a storage device is passive, but when it is loaded into memory and execution begins, it becomes a process.

Ex- When a user opens a web browser, the browser program becomes one or more running processes.

Process States

During its lifetime, a process passess through different states.

i) New State

In the New state, the operating system is creating the process - the required resources and process information are being prepared.

2) Ready State

In the Ready state, the process is present in memory and is waiting for the CPU to execute it.

3) Running State

In the Running state, the CPU is executing the instructions of the process.

4) Waiting / Blocked state.

A process enters the Waiting or Blocked state when it has to wait for an event, such as completion of an input/output operation.

5) Terminated State

When the process completes its execution or is stoped by the operating system, it enters the Terminated state.

Q.5)

→ A university computer laboratory contains multiple computers and is accessed by students, faculty members, lab assistants, and administrators. Proper authentications and authorization mechanisms are required to prevent unauthorized access and protect university resources.

Authentication

Authentication is the process of verifying the identity of a user.

1) User Accounts.

Every student and staff member should have a separate user accounts.
Eg - student01, faculty01, labAdmin.

2) Passwords

Users should be required to use strong passwords. Passwords should not be shared with other users and should be stored by the authentication system using secure password - hashing mechanisms.

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Authorization

Authorization determines what an authenticated user is allowed to access or perform.

Role - Based Access Control

Role	
Student	Use applications and access academic files
Faculty	Access teaching resources and manage course files
Lab Assistant	Perform limited laboratory administration.
IT Administrator	Manage users, systems, and security settings.

Q.4)

→ Protection and security are closely related concepts in a OS, but they have different purposes.

Protection	Security
1) Controls access to system resources	Protects the system from threats
2) Focuses on authorized resource usage	Includes protection against unauthorized access & attacks
3) Uses permissions & access-control mechanisms	Uses authentication, encryption, access control, monitoring, etc.
4) Mainly controls internal access.	Has a broader system-wide focus.

Access Control of System Resources

Read → View contents

Write → Modify contents

Execute → Run the file.

Processes.

The OS restricts users & processes from accessing or controlling processes for which they do not have appropriate privileges.

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Memory

Memory protection prevents one process from accessing another process's protected memory. This helps prevent accidental or unauthorized modification of data belonging to another process.

(Q-5)

→ Authentication is an important security mechanism used by OS to verify the identity of users.

1) Password-Based Authentication

Password authentication uses a combination of a username and password.

2) Multi-Factor Authentication

Multi-Factor Authentication (MFA) requires two or more independent authentication factors.

3) Biometric Authendication

Biometric authentication identifies users using physical characteristics such as: Fingerprints, Facial characteristics & Iris patterns.

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Suitable Approach for a University Environment.

- Biometric authentication can be best & compatible devices as a convenient authentication method, particularly when integrated securely with the OS.

Q.6)

→ A computer may have different types of passwords that operate at different stages of the startup process. The most important distinction is between firmware-level authentication and OS authentication.

1) BIOS/UEFI Password

BIOS or UEFI is firmware that initializes the computer hardware and begins the boot process.

A BIOS/UEFI password can be used to restrict access to firmware settings and, depending on the device configuration, can also restrict boot-related operations.

It is checked during the firmware stage, before normal OS login.

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2) System/login Password

A system or login password is used when the OS displays its login interface. Ex- When a user starts a Windows or Linux computer, they may be asked to enter their username & password.

3) Operating-System Password

An operating-system password is generally the password associated with an OS user account. In many contexts, OS passwords and system/login passwords refer to the same general type of credential.